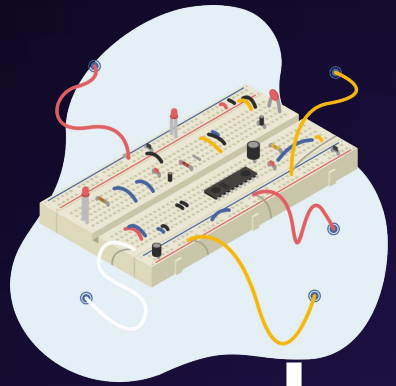
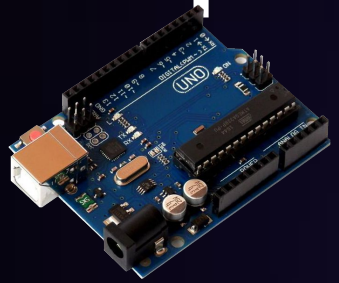
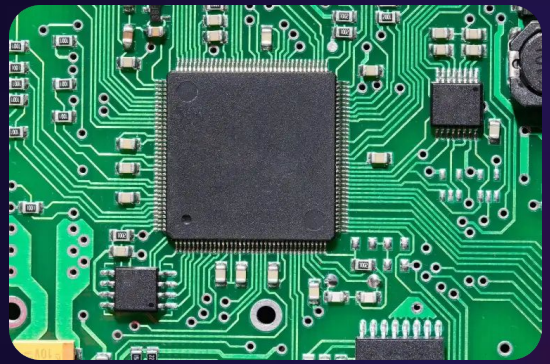
Architecture of electronic and computing machine and computing systems







Course Code: S002
Author: Mirlan Nurbekov

ALA-TOO INTERNATIONAL UNIVERSITY
IT & BUSINESS COLLEGE

Input and output

OUTPUT - Arduino sends a signal (LED, buzzer, motor)

INPUT - Arduino reads a signal (button, sensor, switch)

SWITCH BUTTONS



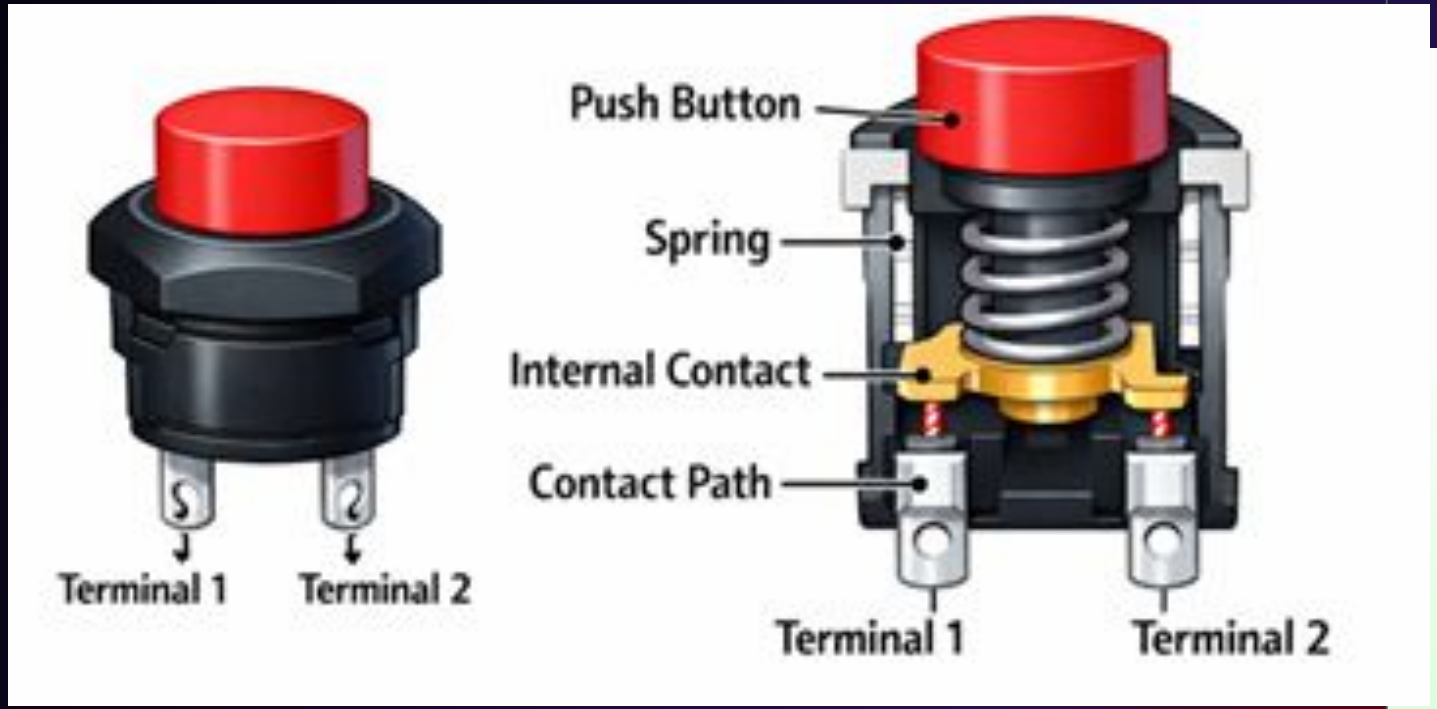




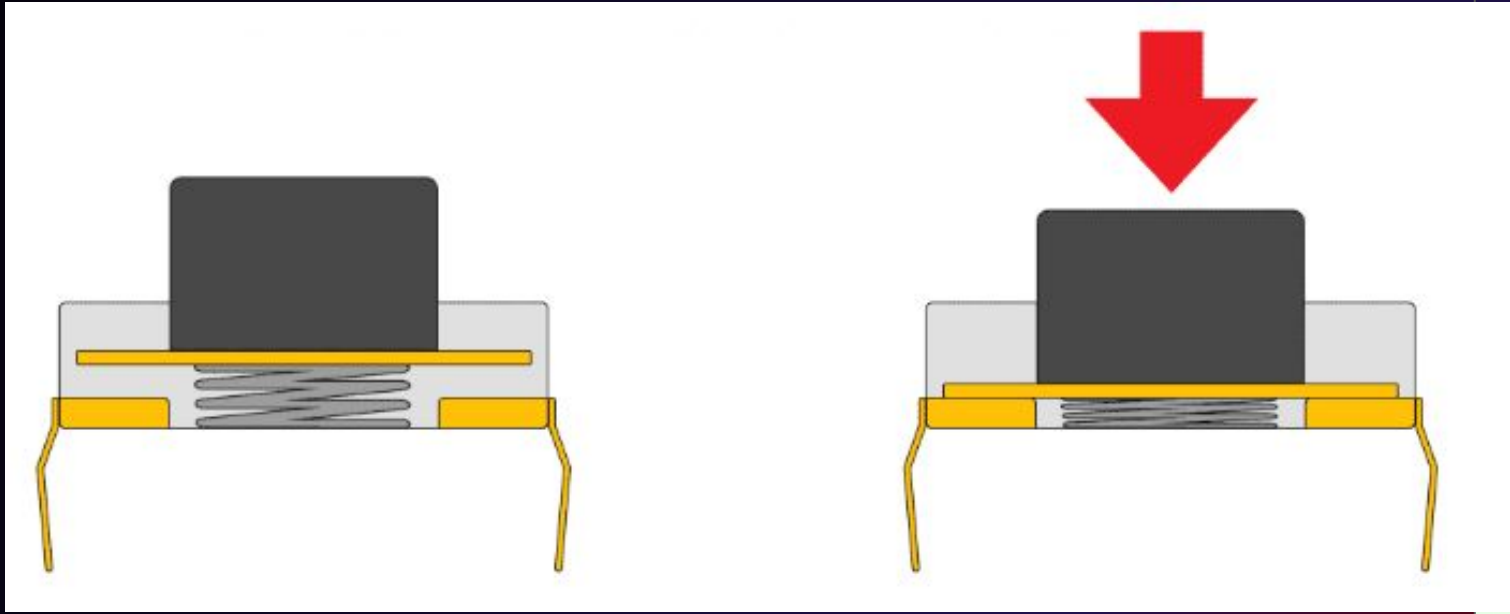
Push button (single pole, single throw)



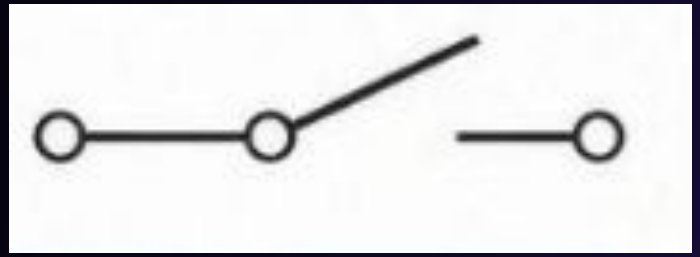
Push button (single pole, single throw)

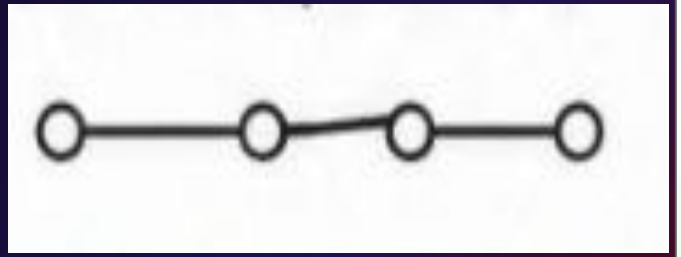


Push button (single pole, single throw)



Push button (single pole, single throw)

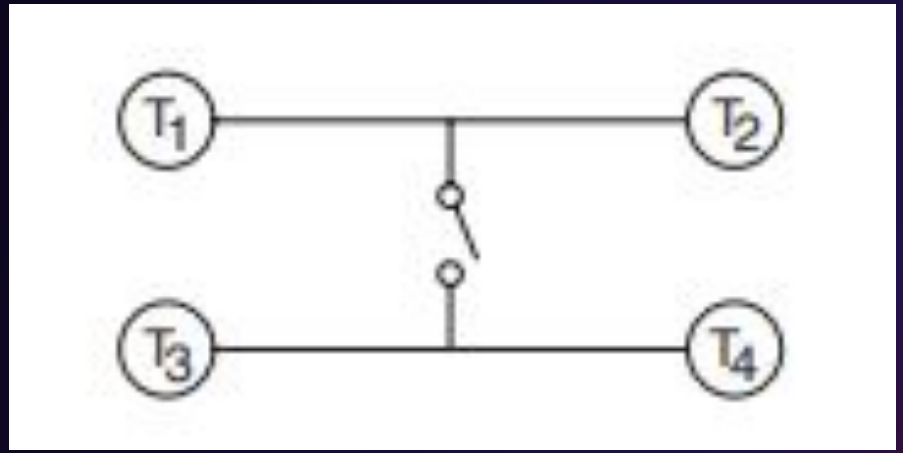




4 pin button (tactile switches)



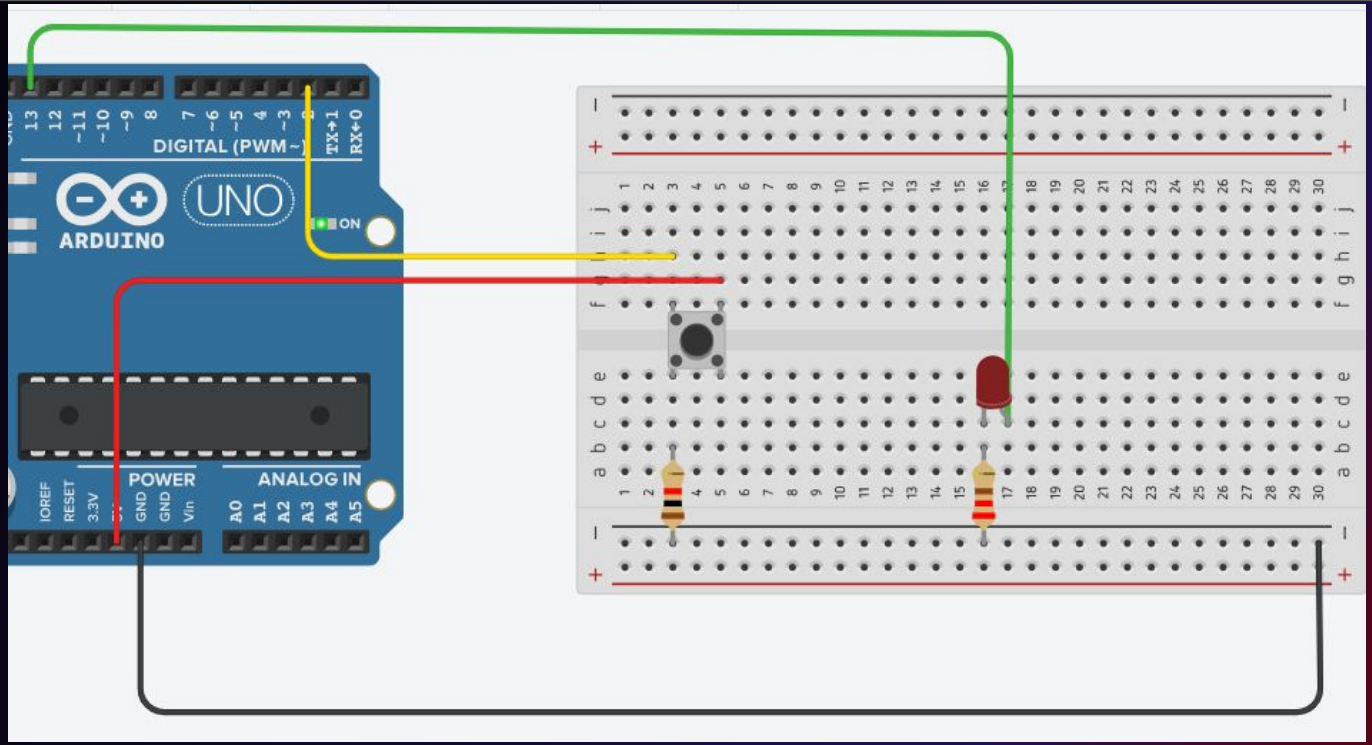
4 pin button (tactile switches)



Task L009

Build the circuit on a breadboard and connect it to the Uno. Write a program so the built-in LED turns ON while the button is pressed and turns OFF when the button is released.

Task L009


DIGITAL (PWM~)
UNO
ARDUINO
ON
POWER
ANALOG IN
IOREF
RESET
3.3V
GND
GND
Vin
A0
A1
A2
A3
A4
A5
TX→1
RX←0

Task L009

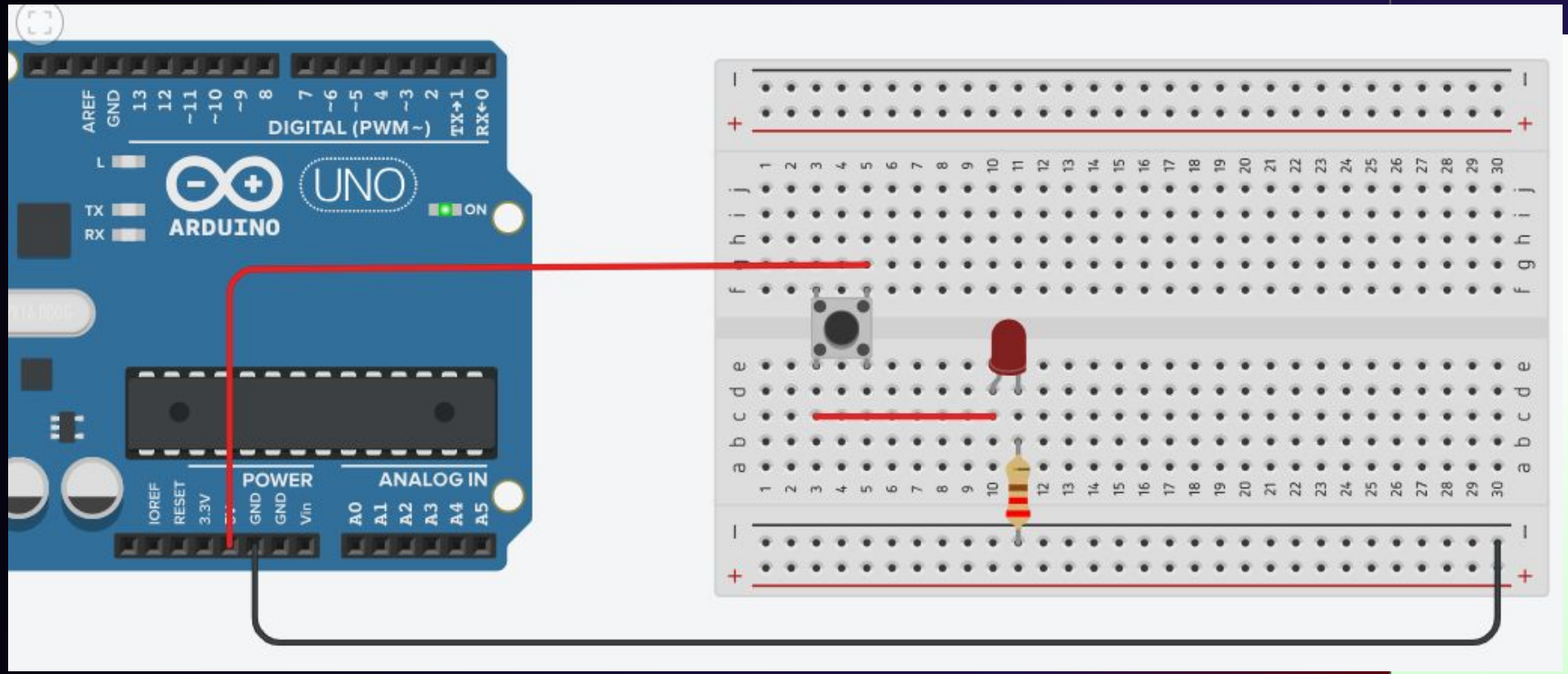
L009.ino

```
1   const int buttonPin = 2;
2   const int ledPin = 13;
3
4   int buttonState = 0; // to read button status
5
6   void setup() {
7     pinMode(ledPin, OUTPUT);
8     pinMode(buttonPin, INPUT);
9   }
10
11  void loop() {
12    buttonState = digitalRead(buttonPin); // reading the value of the button status
13
14    if(buttonState == HIGH)
15    {
16      digitalWrite(ledPin, HIGH);
17    }
18    else{
19      digitalWrite(ledPin, LOW);
20    }
21  }
```

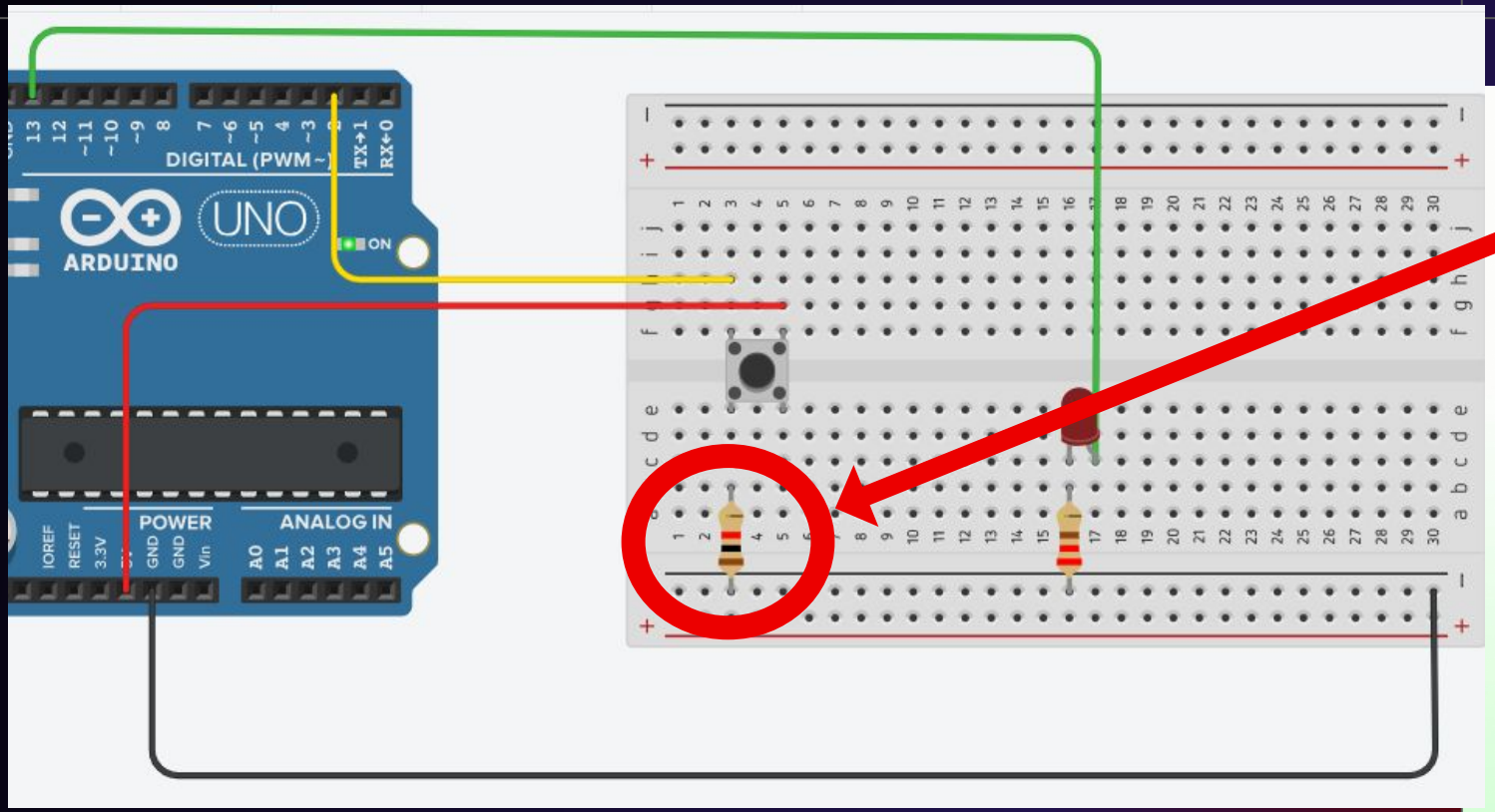
EASIER WAY



Task L009 (ez way)



Why do we have this resistor?



Why do we have this resistor?

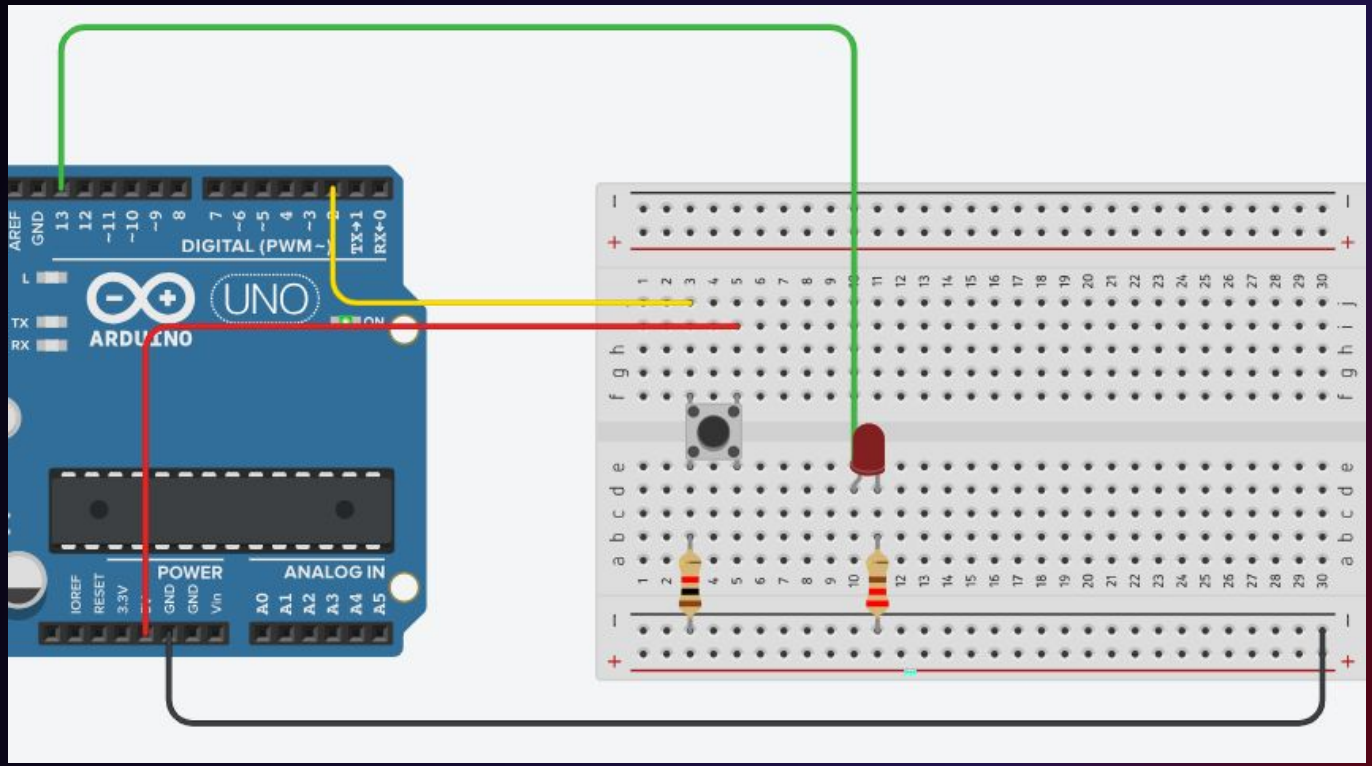
When the button pressed: pin 2 is connected to 5V, so it reads HIGH and the LED turns ON. This part works.

When button released: pin 2 is connected to nothing, so it is floating. It can read HIGH or LOW at random, and it can change if you touch the wire, move your hand near it, or bring a phone close. The LED may flicker, stay ON after you release the button or turn OFF correctly just by luck.

Task L010

Use the circuit from L009. Write a program so that after ONE press of the button the LED blinks 10 times: ON for 500 ms, then OFF for 500 ms each time. After the last blink the LED stays off and the program waits for the next press.

Task L010


AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
UNO
ON
POWER
ANALOG IN
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

Task L010

```
const int buttonPin = 2;
const int ledPin = 13;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(buttonPin, INPUT);
}

void loop() {
  if (digitalRead(buttonPin) == HIGH) {       // button pressed
    for (int i = 0; i < 10; i++) {            // repeat 10 times
      digitalWrite(ledPin, HIGH);             // LED on
      delay(500);                             // wait 500 ms
      digitalWrite(ledPin, LOW);              // LED off
      delay(500);                             // wait 500 ms
    }

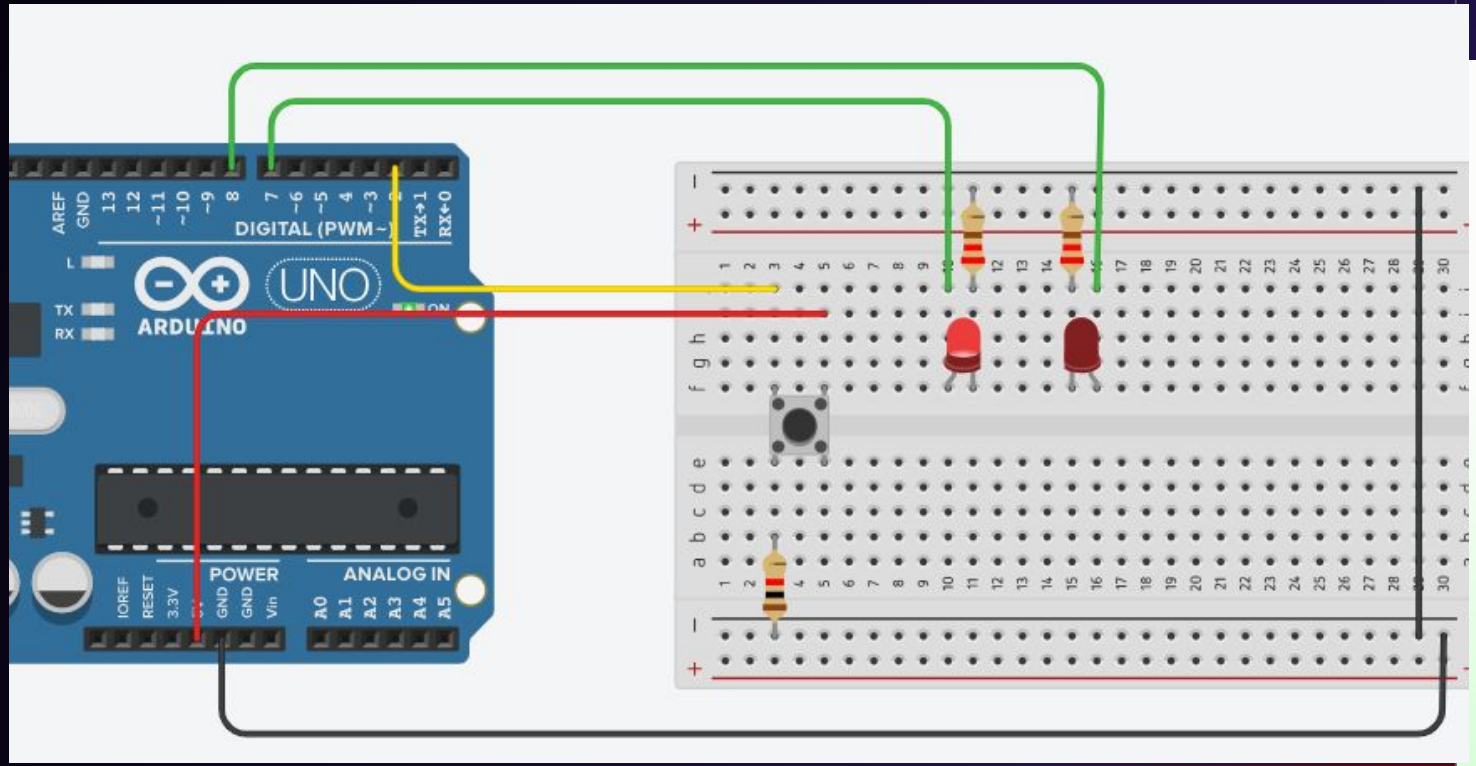
    while (digitalRead(buttonPin) == HIGH) { // wait until button
    }
  }
}

```

Task L011

Build a circuit with two LEDs and one button. Write a program so LED 1 is ON from the start and LED 2 is OFF. Every time the button is pressed, the LEDs switch: LED 1 turns OFF and LED 2 turns ON and on the next press they switch back. The LEDs must never be on at the same time.

Task L011



Task L011

```
const int buttonPin = 2;
const int led1Pin = 8;
const int led2Pin = 7;

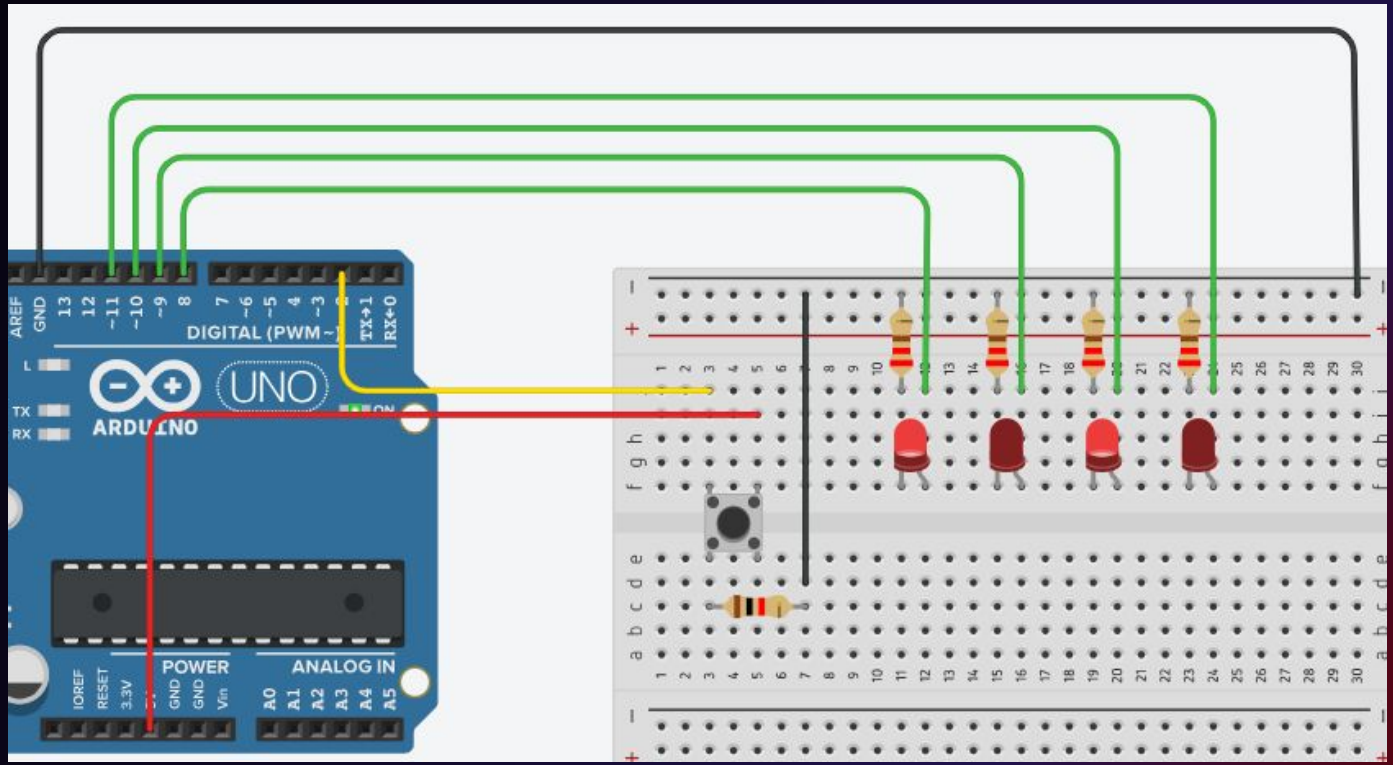
bool led1On = true;           // LED is ON
bool lastButton = LOW;        // button state

void setup() {
  pinMode(buttonPin, INPUT);
  pinMode(led1Pin, OUTPUT);
  pinMode(led2Pin, OUTPUT);
  digitalWrite(led1Pin, HIGH);    // LED 1 ON from the start
  digitalWrite(led2Pin, LOW);     // LED 2 OFF
}
void loop() {
  bool button = digitalRead(buttonPin);
  if (button == HIGH && lastButton == LOW) {    // button was pressed
    led1On = !led1On;                           // switch the state
    digitalWrite(led1Pin, led1On);
    digitalWrite(led2Pin, !led1On);
    delay(50);                                  // ignore bounce
  }
  lastButton = button;
}
```

Task L012

Build a circuit with four LEDs and one button. Write a program so all LEDs are OFF at the start. After ONE press of the button the LEDs show the number 10 in binary (1010): LED 1 is ON, LED 2 is OFF, LED 3 is ON, LED 4 is OFF. The LEDs stay in this state after the button is released.

Task L012



Task L012

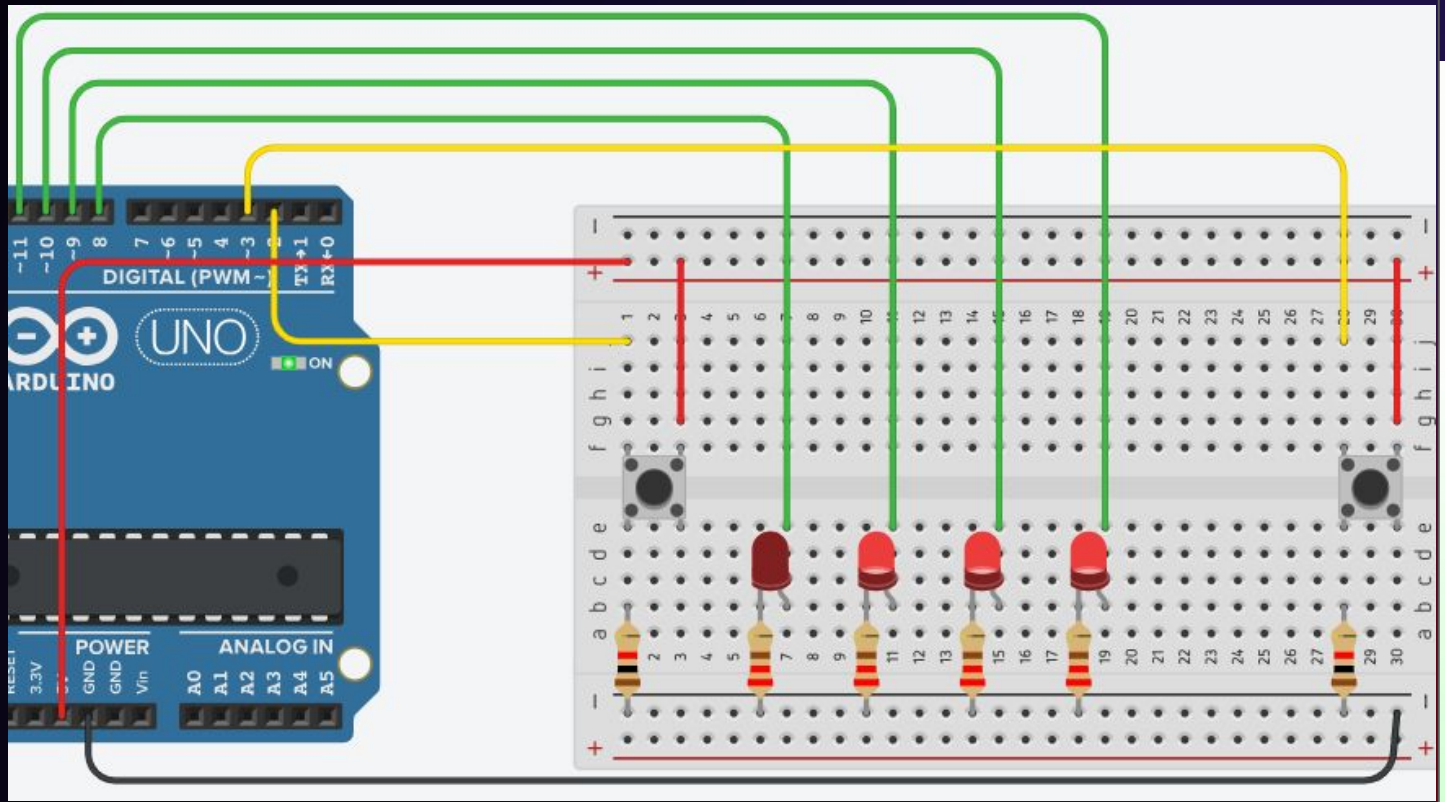
```
const int buttonPin = 2;
const int led1Pin = 8;
const int led2Pin = 9;
const int led3Pin = 10;
const int led4Pin = 11;

void setup() {
  pinMode(buttonPin, INPUT);
  pinMode(led1Pin, OUTPUT);
  pinMode(led2Pin, OUTPUT);
  pinMode(led3Pin, OUTPUT);
  pinMode(led4Pin, OUTPUT);
}
void loop() {
  if (digitalRead(buttonPin) == HIGH) {
    int number = 10;
    digitalWrite(led1Pin, bitRead(number, 3));
    digitalWrite(led2Pin, bitRead(number, 2));
    digitalWrite(led3Pin, bitRead(number, 1));
    digitalWrite(led4Pin, bitRead(number, 0));
  }
}
```

Task L013

Build a circuit with four LEDs and two buttons. Write a program so all LEDs are OFF at the start. When button 1 is pressed, the LEDs show the number 10 in binary (1010). When button 2 is pressed, the LEDs show the number 7 in binary (0111). The LEDs keep the last number until the other button is pressed, so the program can switch between 10 and 7 as many times as needed.

Task L013



Task L013

```
const int button1Pin = 2;
const int button2Pin = 3;
const int led1Pin = 8;
const int led2Pin = 9;
const int led3Pin = 10;
const int led4Pin = 11;

void showNumber(int number) {
  digitalWrite(led1Pin, bitRead(number, 3));
  digitalWrite(led2Pin, bitRead(number, 2));
  digitalWrite(led3Pin, bitRead(number, 1));
  digitalWrite(led4Pin, bitRead(number, 0));
}
void setup() {
  pinMode(button1Pin, INPUT);
  pinMode(button2Pin, INPUT);
  pinMode(led1Pin, OUTPUT);
  pinMode(led2Pin, OUTPUT);
  pinMode(led3Pin, OUTPUT);
  pinMode(led4Pin, OUTPUT);
}
void loop() {
  if (digitalRead(button1Pin) == HIGH) {
    showNumber(10);}
  if (digitalRead(button2Pin) == HIGH) {
    showNumber(7);
  }
}
```

Creative pulse



Creative pulse

Invent an idea that uses buttons and LEDs and then build the circuit + write the program.
Your idea must be something that someone could really use, or something fun to play.

Self test mcq

1. What does the resistor between pin 2 and GND do in the button circuit?

A) Limits the current going through the LED
B) Protects the button from wearing out quickly
C) Keeps pin 2 LOW when the button is released
D) Lets the Arduino read the button without pinMode()

Self test mcq

2. In L009, the button is pressed. What does digitalRead(buttonPin) return?

A) HIGH
B) LOW
C) The pin number 2
D) The time the button was held

Self test mcq

3. In L010, what is the job of the while loop after the for loop?

A) It makes the LED blink ten more times
B) It turns the LED off after the last blink
C) It repeats the blinking until a reset
D) It waits until the button is released

Self test mcq

4. In L011, why do we use the lastButton variable?

A) To count how many times the LEDs have switched
B) To react once per press, not on every loop
C) To set the delay time after each press
D) To choose which LED starts ON

Self test mcq

4. In L011, why do we use the lastButton variable?

A) To count how many times the LEDs have switched
B) To react once per press, not on every loop
C) To set the delay time after each press
D) To choose which LED starts ON

Self test mcq

5. In L013, which LEDs are ON after showNumber(7)?

A) LEDs 1 and 2
B) LEDs 1, 2 and 3 only
C) LEDs 2, 3 and 4
D) All four LEDs

Answers

1. C
2. A
3. D
4. B
5. C

Home work

Build a circuit in Tinkercad using resistors, LEDs, push buttons and one component you chose last week. Use as many Arduino Uno pins as you can, but every part must have a job. Your project should be a device that solves a problem or a game people would enjoy, not random components wired together. The program must react to the buttons. For the next class, get your simulation ready and be prepared to show it in 2 minutes: what the idea is, how it works, how many pins you used and what was the hardest part.